



Toward Event-Based Noise-Robust High Density Particle Velocimetry

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The necessity of understanding and analyzing the flow field characteristics has prompted researchers to strive for more precise tools and systems to enhance the accuracy of flow velocity measurements. We propose in this paper the noise-robust Particle Event Velocimetry (PEV), which is a particle-level estimate of the flow field based on the analysis of individual seed particles observed and tracked using event-based camera. Compared to conventional frame-based (FB) Particle Image/Tracking Velocimetry (PIV/PTV) and event-based (EB) particle velocimetry methods, the proposed PEV's flow field achieves higher spatial resolution limited only by the density of the seeded particles while maintaining experimental simplicity. We also developed an event-based camera simulator for synthetic data with a ground-truth motion field to objectively benchmark the proposed PEV against other FB and EB methods and analyzed these competing methodologies' ability to reconstruct sharp motion boundaries and flow field curvatures. The results include real-world event-based camera data experiments, in addition to the benchmark results which show that the algorithm developed for PEV was able to achieve a superior velocimetry compared to other algorithms.

I. Nomenclature

PEV	=	Particle Event Velocimetry
PIV	=	Particle Image Velocimetry
PTV	=	Particle Tracking Velocimetry
EB	=	Event-based
FB	=	Frame-based
FOV	=	Field of View
2D-2C	=	Two-dimensional Two-component
IE	=	Inceptive event
TE	=	Trailing event
PL	=	Pulse filter
PS	=	Positive filter
$\hat{V}_x, \hat{V}_y$	=	Velocity in x and y direction
$\vec{A}c_x, \vec{A}y_x$	=	Acceleration in x and y direction
x, y	=	Event location in x and y per pixels
T_x, T_y	=	Track center
$\hat{x}, \hat{y}$	=	Line-fitted track location

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II. Introduction

UNDERSTANDING the characteristics of the flow around objects plays an important role during the development of energy efficient and structured robust aerodynamic designs. Owing to recent technological advancements, state-of-the-art Particle Image Velocimetry (PIV) and Particle Tracking Velocimetry (PTV) systems have achieved exceptional capabilities to estimate the flow velocity using complex setups and cutting-edge hardware. Meanwhile, classical two-dimensional two-component (2D-2C) setups remain a powerful and popular tool for analyzing two-dimensional complex flow phenomena due to its robust results, experimental simplicity, and validated algorithms [1, 2].

To estimate a flow velocity using the double exposure 2D-2C Frame-Based (FB) setup, a single high-speed camera is used to observe the seeded particles moving within the flow medium, such as a gas or liquid, while crossing the illuminated field of view (FOV). By capturing the movement of the seeding particles and estimate its velocity, the flow velocity can be estimated relatively using non-intrusive approach. After recording two consecutive frames, the PIV [3] is performed by estimating the average movement of the particles within a spatial window (sub-interrogation areas) using cross-correlation to find the best spatial window match across the two frames.

Alternatively, the PTV [4] estimates the velocity of the individual particles by tracking the trajectory of a particle between two frames using particle matching techniques. Even though the PIV is more commonly used for its robustness against the noise and require less fancier hardware, the drawbacks for both methods include difficulty in analyzing slow and fast motions simultaneously, limits on frame rate/temporal resolution, high data throughput, high computational complexity, and high camera hardware costs associated with high-speed imaging [1, 5]. In addition, the experimental hardware cost and complexity increases as the desired accuracy increase, as well as the computational and storage costs stemming from large data volume.

Thanks to the state-of-the-art event-based cameras boasting microsecond-order resolution, it possible to observe a fast moving phenomena with a significant reduction in the data throughput without sacrificing the temporal resolution. The reduction comes from the fact that the EB-sensor produces events only when a change of intensity occur for each pixel individually while the FB systems capture a full frame resolution at selected frequency (as a *frame*), as demonstrated in Figure 2. Therefore, our goal is to use this emerging sensor technology to develop a noise-robust particle velocity measurement system with a higher temporal accuracy while maintaining the experimental simplicity and costs.

In this work, we propose in this paper the noise-robust Particle Event Velocimetry (PEV), a new method of particle-level estimate of the flow field based on the analysis of individual seed particles observed and tracked using event-based camera. We also developed an event-based camera simulator for synthetic data with a ground-truth motion field to objectively benchmark the proposed PEV against other FB and EB methods and analyzed these competing methodologies' ability to reconstruct sharp motion boundaries and flow field curvatures. Both synthetic as well as real-world event-based camera data experiment suggest that the proposed PEV was able to achieve a superior velocimetry compared to existing FB PIV methods, as well as existing EB techniques.

A. Event Camera

Since 1992 when Mahowald introduced dynamic vision sensing (event-based sensing) to mimic the function of the human retina, image sensor technology has improved significantly. 2008 marked first commercially available event-based cameras by Inivation [6], and Prophesee recently developed backside illuminated megapixel sensor in 2021. Thanks to its low power, high dynamic range, and high temporal resolution, event-based image sensors have the

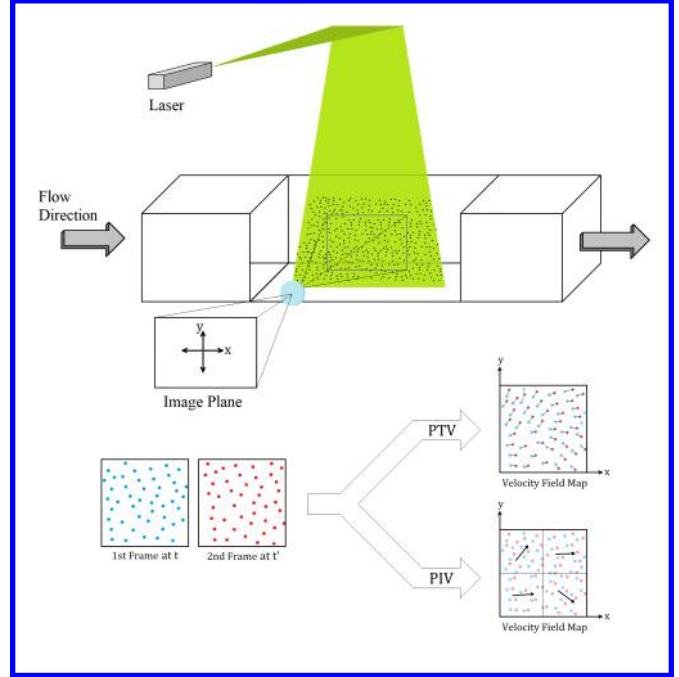


Fig. 1 Classical 2D-2C frame-based PIV/PTV setup.

potential to positively impact many applications [7–13].

Unlike the conventional framing cameras, each pixel on the event camera operates independently and asynchronously. Instead of synchronously capturing the accumulation of photons to yield frames, each pixel reports an *event* only if there is a change of intensity with time resolution of microseconds regardless of what is happening to the neighboring pixels. It produces a positive event if the intensity increased within a certain threshold and a negative one if it was otherwise. Thus, the event camera will generate asynchronous data that include the special location of the event, the time stamp, and the polarity of the event. Unlike a framing camera, the event camera theoretically will not produce any event unless there is change of intensity that exceeds the threshold. Thus it is particularly advantageous for observing sparse high speed phenomenon (such as particle velocimetry), with savings in data throughput and computational complexity. See illustration in Fig. 2.

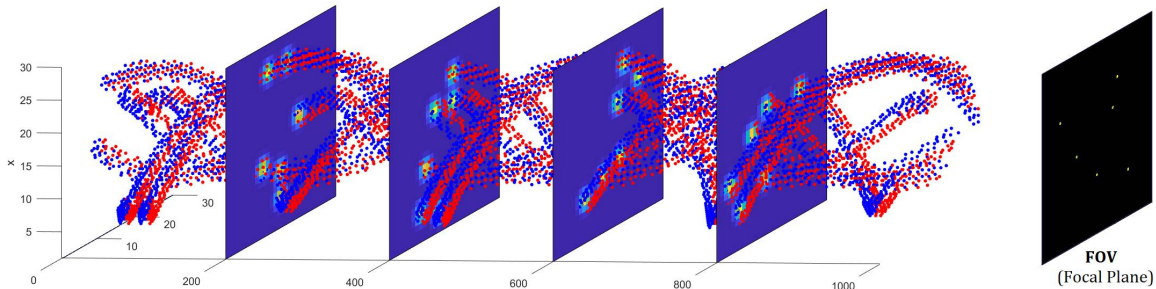


Fig. 2 Comparison between the output of the conventional camera (Frames) vs event-based camera (Events).

Events generated from asynchronous pixels require a very different kind of processing than the cross-correlation approaches used in the conventional PIV/PTV for framing cameras. Although some techniques can be borrowed with some modifications (like tracking algorithms), we develop below event-based processing techniques that makes full considerations for the asynchronous nature of the event streams.

B. Literature Review

As the resolution and capability of the event-based sensor improved, researchers began exploring its application to imaging high-speed phenomenon. Its application to particle velocimetry is summarized in Fig.3.

Drazen in 2011 [14] provided the first proof of concept for event-based particle velocity. Years later, Wang in 2020 [15] proposed the stereo event-based particle tracking velocimetry based on 2D optical flow. To improve robustness and overcome the uncertainty of optical flow-based tracking and matching, the output of this system is an average velocity per space-time volume using two event cameras.

More recently, Willert in 2022 [16] estimated the velocity of the fluid in 2D by finding the average velocity per voxel (x, y, t) using maximum contrast as an alternative to the conventional cross-correlation-based PIV. Although averaging of vector velocity was shown to improve robustness for a dataset with very large integration time (10mSec), the quality would deteriorate in presence of strong gradients in flow field.

	Frame Based	Event Based
Per Particle	PTV	Drazen (2011) Wang (2020) PEV (ours)
Per Window Voxel	PIV	Willert (2022)

Fig. 3 Comparison of PTV[4], PIV[3], Drazen (2011)[14], Wang (2020)[15], and Willert (2022)[16] against PEV based on estimation method and category.

III. System Design

The proposed Particle Event Velocimetry (PEV) system aims to improve the event-based velocity estimation in terms of noise robustness, the accuracy of the estimated spatial-temporal particle trajectory, and the tolerance to very high particle density. This is accomplished by exploiting specific characteristics of event-based sensors to overcome uncertainties in the event stream, and by estimating the particle trajectory across tracked events using nonlinear fitting at microsecond resolution. Because the flow field is computed from individual tracks without spatial-temporal windowing, the resolution of the PEV is largely determined by the density of the seeded particles. Since the event camera produces asynchronous events, the algorithm design will have a different approach than the FB methods to handle the raw data before performing any analysis. The resultant PEV flow field is able to preserve the motion boundaries with higher spatial-temporal fidelity.

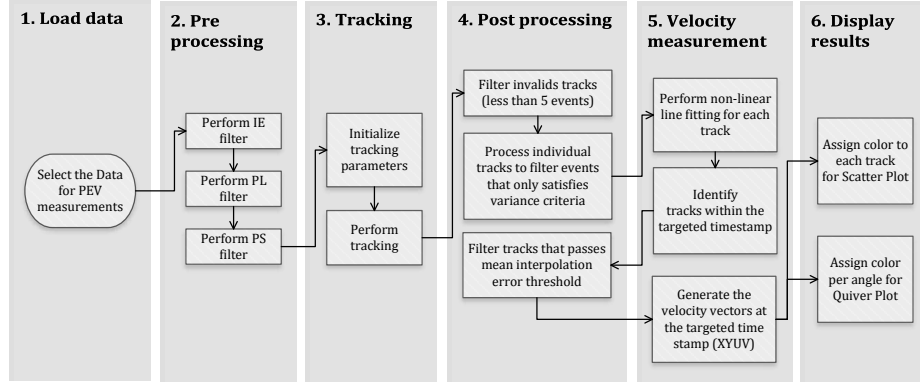


Fig. 4 Flow chart of the PEV system including all stages.

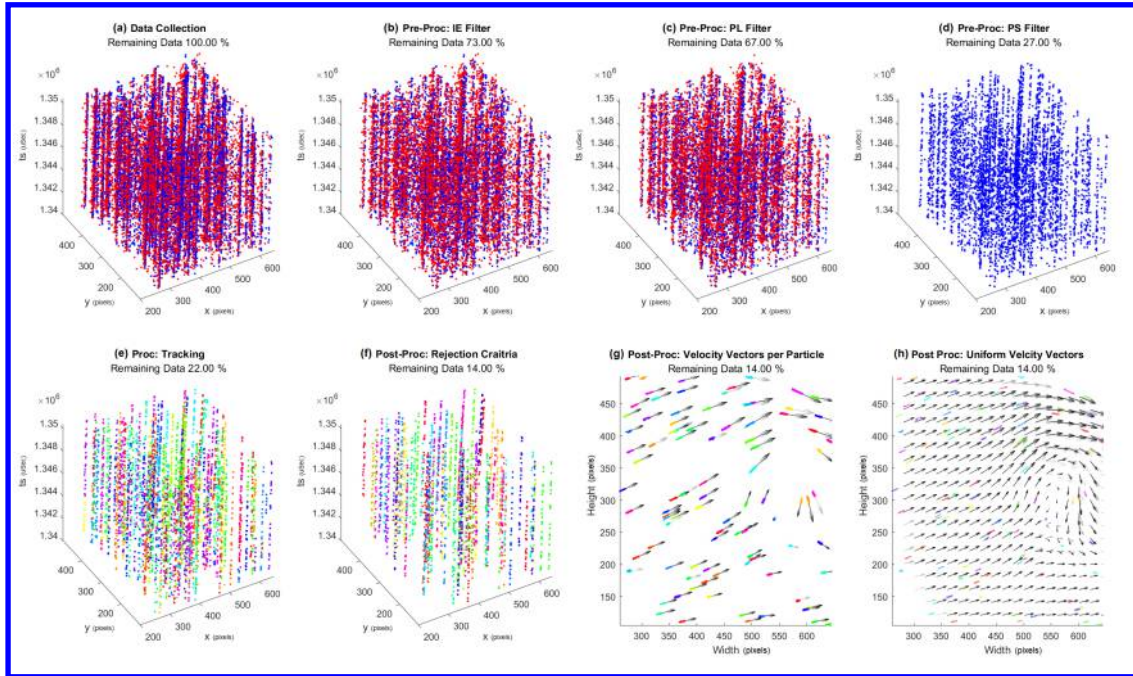


Fig. 5 The proposed PEV system overview.

Fig. 4 shows the system diagram of our proposed PEV, starting from the raw data to the estimation of velocity vectors, details of which are explained in the subsequent subsections. As shown by the examples in Fig. 5 illustrates

the processed events at each stage, the system extracts the salient events from the noisy raw data at the pre-processing stage, partition events into multiple tracks each corresponding to a particle, and then finally perform velocity vector measurements at the particle level. We use the Inceptive Event (IE) filter which will reduce the size of the data and keep the events with most accurate timing [17]. The Pulse (PL) filter will be used to minimize the noisy and uncertain events by finding the reliable pairs of positive and negative events corresponding to small moving particles by temporal thresholding. The last step at the pre-processing stage is the Positive events (PS) filter which keeps only the positive events they correspond to the exact timing of the particle arrival times. In the real world example shown in Fig. 5, we gain significant robustness to noise while reducing the data volume to a third of the raw event stream.

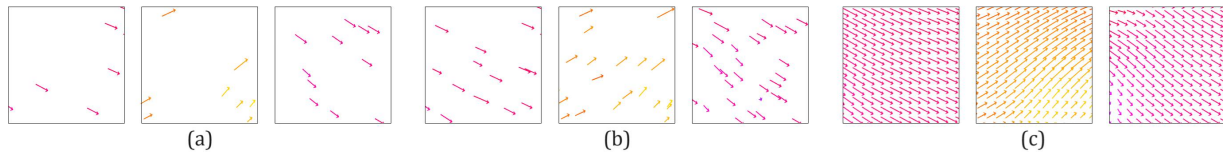


Fig. 6 Velocity vectors of the experimental data for: (a) only active tracks at the exact timestamp, (b) with tracks available within a short-time window, (c) with interpolated velocity vectors using the neighbor vectors to fill any gaps.

Finally, the data of each track is analyzed to yield a velocity vector per particle while using a rejection criteria at the post-processing stage. If the flow field is static, the system can use old tracks to predict the velocity at less dense area. If dynamic, we use interpolation tools to fill the gaps based on the neighboring measurement as shown in Fig. 6.

A. Pre-processing

The main goal at the pre-processing stage is to reduce the noise and extract salient events that will not only contribute to the quality of the analysis, but reduce the data volume dramatically before performing the tracking and line fitting analysis. Pre-processing is comprised of three filters: Inceptive event filtering (IE), pulse filter (PL), and positive event filter (PS)

1. Inceptive Event Filter (IE)

Theoretically, the particle velocimetry experiment is conducted in dark environment where the camera sensor receives photons reflected from seeding particle intersecting the laser sheet. The event pixel sensor operates as intensity change detector with pre-defined threshold, recording the exact timing of the particle arriving at a pixel.

In practice, a single event-based pixel typically yields multiple events at a given pixel due to the fact that the log-intensity change is greater than a single threshold value. Due to the large contrast between the dark background and the laser illuminated seed particle, this will lead to data redundancy. Even more serious problem is that these multiple events are read out sequentially, introducing an ambiguity in time due to latency. Additionally, there are a high number of noisy events that do not correspond to a meaningful changes in intensity.

Willert [18] attempted to handle this issue by proposing discontinuous laser. However, it sacrifices the microsecond resolution of the event-based camera, effectively turning the event stream to act like a conventional framing camera. It shares disadvantages of the conventional PIV and PTV, including the loss of spatial-temporal resolutions stemming from spatial windowing and correlation methods. In addition, laser pulsing causes multiple pixels to generate events simultaneously, overwhelming the readout circuit and thus severely limiting the seed particle density that the event-based particle velocimetry can handle.

In our work, we propose to leverage the Inceptive Event filter, designed by Bladwin *et al.* [17] to address the data redundancy and timing ambiguity issues. Among multiple events occurring in a series, the initial event (called inceptive event or IE) corresponds to the precise moment of the particle encountering a particular pixel. The subsequent events (called trailing event or TE) experience latency, due to a cycle of resets, refractory period, and read out circuit delays in 2-200 microsecond ranges [19]. Thus we consider TE events to be nuisance and redundant.

The process of the IE filter labeling each event as “IE” or “TE” is shown in the flowchart in Fig. 7 (left). The subplots on the right illustrates this process for the positive (blue) and negative (red) events, and the result of removing the TEs. The remaining IE positive and negative events correspond unambiguously to the accurate spatial-temporal timing of the particles arriving and exiting the pixel, respectively.

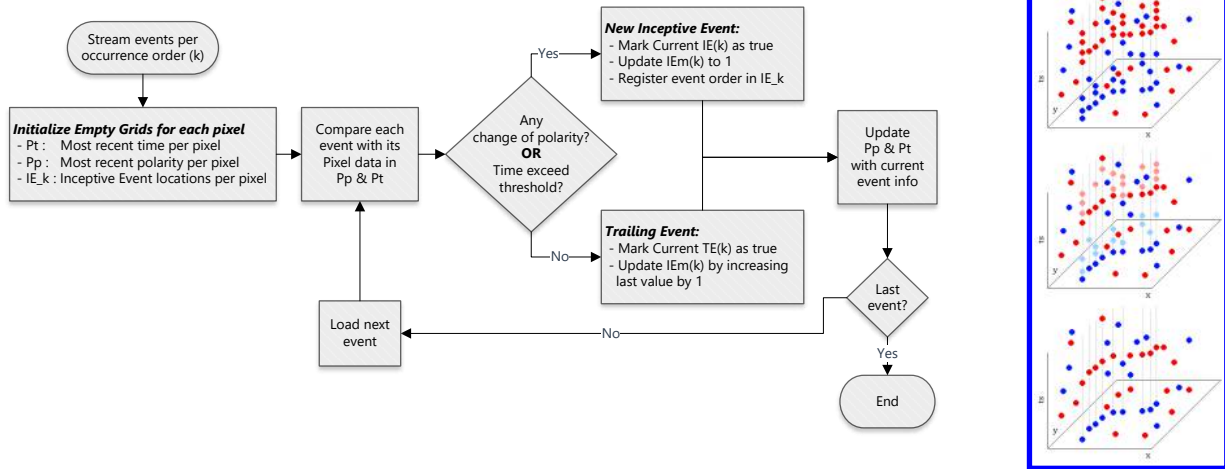


Fig. 7 Flow chart of Inceptive event filter function (left) and demonstration of the events before and after performing the IE filter (right).

2. Pulse Filter (PL)

The pulse filter (PL) aims to reduce the noisy events to enhance the accuracy of tracking and velocity measurements. We consider the positive event followed by negative event within certain time frame corresponds to describe a legitimate particle movement. This is because event-based pixel is observing an illuminated particle against a dark background and it will report leading positive event upon the arrival of the particle at a particular pixel, followed by negative ones when it leaves since the intensity decreases (like a pulse signal).

Thus, the pulse filter is designed to find a pair of positive and negative events occurring within a certain temporal threshold. The operation of this process is explained in Fig. 8 (left) while Fig. 8 (right) shows a visualization of sparse event data, and how the data have less noise after the PL filter. While PL is effective at removing noisy events, there are some cases where the events are missing due to noise. To safeguard against this, we also consider looking at neighboring pixels to allow the pulse timing to be calculated.

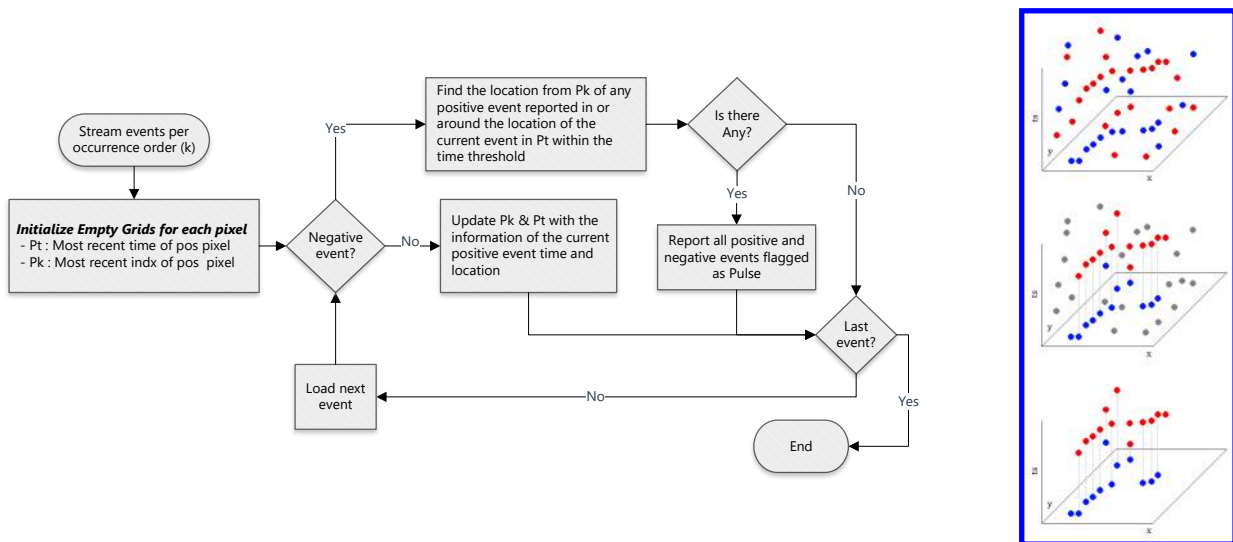


Fig. 8 Flow chart of the Pulse filter algorithm function (left) and demonstration of the events before and after performing the PL filter (right).

3. Positive Event Filter (PS)

The output of the PL filter comes in pairs of positive and negative events. Although tracking can be performed on either positive or negative events, positive events are preferred because it corresponds to the arrival timing of the seed particle. The negative events in some cases can be affected by the small delay stemming from the sequential reporting of the trailing events. For this reason, at the positive event filter (PS) stage, we remove the negative events.

B. PEV Processing

After the pre-processing step that reduced data redundancy, removed noisy events, and extracted the most salient events, we are ready to perform particle event velocimetry (PEV). The processing is comprised of three major steps. First, events are tracked and analyzed. Second, after each event is track, we perform a second degree polynomial line-fitting, which allows us to robustly estimate the velocity and project it to every observed particle. Finally, we perform a post-processing rejection of invalid tracks. The details of these steps are described in the following subsections. Generally we take the “track-before-detect” approach where we take the track quality into the consideration of whether the flow field estimates are valid.

1. Weighted Tracking

Prior to velocity estimation, each event must be assigned to the particle track it belongs to. Since the events are processed in the order it was generated, the first event will be declared as the first track. To decide whether the following event belongs to the same track or not, the track error must be computed between the current event and all tracks using the Euclidean distance:

$$\text{TrackError} = \sqrt{(x - T_x)^2 + (y - T_y)^2}, \quad (1)$$

where (T_x, T_y) is the center of an active track and (x, y) is the location of the event under investigation.

When a noisy event creates a false track within the neighborhood of another valid track, the new event might be closer to the false one which leads to incorrect assignment. To overcome this issue, track assignment should be preferential towards valid tracks with more events. Empirically, we found that weighting scheme relative to the number of events of each track to be effective:

$$\text{TrackWeight} = 1 + \frac{1}{k} \min(\{\# \text{ of events in the track} \}, k), \quad (2)$$

where k is a user parameter with the upper limit to number of events. Notice that this weight is restricted to the range between 1 and 2.

We combine the TrackError and the TrackWeight to arrive at the final TrackScore, as follows:

$$\text{TrackScore} = \text{TrackWeight} \cdot \left(\frac{\text{Track Error}^2}{r^2} - 1 \right), \quad (3)$$

where r is a user-specified radius threshold from the track center. Notice that this score is designed to yield a negative value for $\text{TrackError} < r$ and smaller TrackError indicates a better fit.

Track assignment is made in the following manner. For each new event, we compute the TrackScore for every existing *active* tracks. The track with minimum TrackScore is considered the candidate for assignment. If the minimum TrackScore is negative, the candidate is accepted and the event is assigned to the corresponding track. However, if the score is a positive value, then event is not sufficiently close enough to the track and therefore a new track is formed. Finally, if a track has not had a new event assignment for more than Δ seconds, then it will be labeled as *inactive* track and will not be included in the track assignment considerations for the subsequent events.

2. Velocity Estimation

Owing to the fact that derivatives are sensitive to noise, we propose to perform regression to smooth out the tracks before estimating the flow field. In our work, we used second degree polynomial fitting to account for the spatial-temporally nonlinear particle movement (we found empirically that linear fitting was inadequate). The polynomial

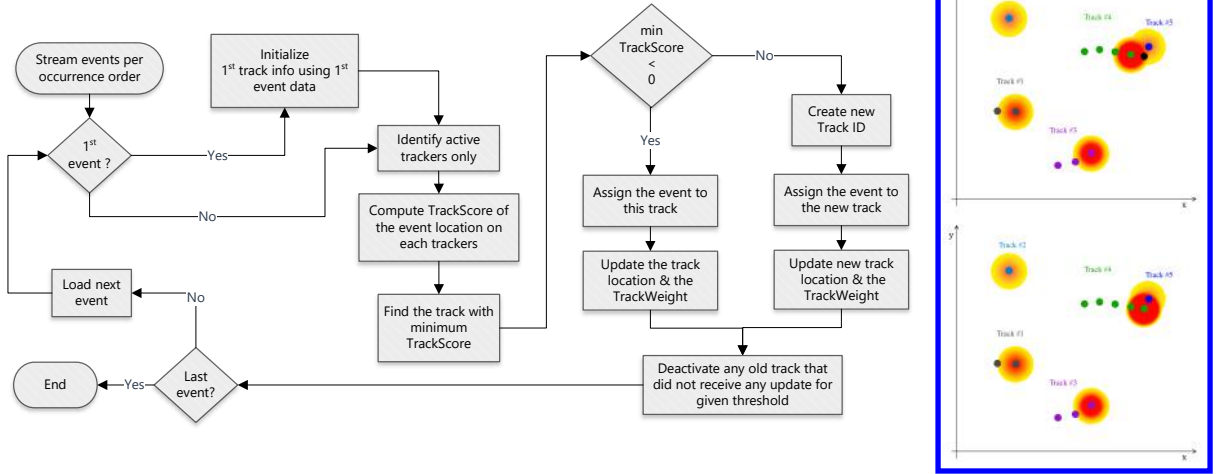


Fig. 9 Weighted tracker flowchart (left) and Weighted tracker demonstration while processing a streaming events (right). As shown, when a new event processed (black dot), it will be assigned to the minimum TrackScore (which is Track 4) not based on the distance which would be in favor of (Track 5). The intensity of the red color refer to the lower TrackScore of the track which is determined by the TrackWeight factor.

regression to three dimensional data $\{(x_n, y_n, t_n)\}$ assumes the following overdetermined model:

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} t_1^2 & t_1 & 1 \\ \vdots & \vdots & \vdots \\ t_n^2 & t_n & 1 \end{bmatrix} \begin{bmatrix} a_x \\ b_x \\ c_x \end{bmatrix}, \quad \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} t_1^2 & t_1 & 1 \\ \vdots & \vdots & \vdots \\ t_n^2 & t_n & 1 \end{bmatrix} \begin{bmatrix} a_y \\ b_y \\ c_y \end{bmatrix}. \quad (4)$$

We use pseudo-inverse to estimate the coefficients $\vec{A}_x$ and $\vec{A}_y$:

$$\vec{A}_x = (M^T M)^{-1} M^T \vec{X}, \quad \vec{A}_y = (M^T M)^{-1} M^T \vec{Y}, \quad (5)$$

yielding a track regression:

$$\hat{X} = M(M^T M)^{-1} M^T \vec{X}, \quad \hat{Y} = M(M^T M)^{-1} M^T \vec{Y}. \quad (6)$$

We may now estimate the velocity at particle (x_n, y_n, t_n) by differentiating the tracks in (4):

$$\hat{V}_x = \frac{\partial}{\partial t} M \vec{A}_x = \begin{bmatrix} 2t_1 & 1 \\ \vdots & \vdots \\ 2t_n & 1 \end{bmatrix} \begin{bmatrix} a_x \\ b_x \end{bmatrix}, \quad \hat{V}_y = \frac{\partial}{\partial t} M \vec{A}_y = \begin{bmatrix} 2t_1 & 1 \\ \vdots & \vdots \\ 2t_n & 1 \end{bmatrix} \begin{bmatrix} a_y \\ b_y \end{bmatrix}, \quad (7)$$

as well as the acceleration: $(2a_x, 2a_y)$.

C. Post Processing: Rejection Criteria

Even though the processed event stream is filtered in multiple stages, it is still possible that noisy events are still present. We propose to apply a post-tracking filter based on track quality. We define the mean squared regression error (MSE) as follows:

$$\text{Mean Squared Error} = \frac{\sqrt{(\hat{X} - \vec{X})^2 + (\hat{Y} - \vec{Y})^2}}{n}. \quad (8)$$

The regression in (6) should minimize the mean squared error between the raw events and its polynomial regression. Thus a high mean squared error in (8) is an indication of a poor track quality. Thus we threshold regression MSE to eliminate tracks that are less reliable.

An additional step is needed to account for the particle tracks that intersects each other that results in a high velocity magnitude error. We adopt a linear change point analysis called PELT [20], that is designed to perform a low computational cost accurate detection for unknown distribution property. The assumption here that if the events on the same track belong to same particle, the timestamp will be reported in approximately a consistent manner. Therefore, by analyzing the time increments of the sequenced events, it can detect if there is any significant changes occurred. The filter selects the longest sequence of events that fits this condition and discard all remaining event from the track. If the largest sequence is equal or larger than threshold, it will be consider as a valid track and it will flagged as pass.

IV. Benchmark

We benchmark the accuracy of the proposed PEV against the prior art in FB and EB methods objectively and subjectively. For objective benchmarking, we developed an event simulator for event-based PIV/PTV applications. Although there are a handful of event-based camera simulators available today[21, 22], they are frame based and not well suited for unresolved targets that resembles moving point sources. We instead model a continuous movement of the point source-like features of seed particles explicitly, along with the optical transfer function and circuit bandwidth in the event-based sensor to yield event streams in a continuous time space based on the flow field. Because the ground-truth flow field is known in simulation, the flow field error can be objectively quantified.

A. Simulation Experiment Setup

Our synthetic model is designed to produce continuous time event streams as well as FB images to be benchmarked against conventional PTV and PIV methods. We derive the changes of intensity of traversing particles at the pixel level in analytical form, and combine it with the pixel simulator that modeling the optical transfer function and circuit bandwidth in the event-based sensor in [6] from observing the intensity changes at the photodiode until reporting the event when occurred.

Specifically, we model each particle as a small Guassian function. Temporally, the intensity profile at a particular pixel of a particle traveling across the detector is a Guassian function, whose width and height are determined by the proximity of the pixel to the particle path. The event based pixel observing this would apply a log function followed by a one pole lowpass filter to model the internal analog signal path. The events are then generated at level crossings (contrast thresholds). See Fig. 10.

Since the Gaussian function and one pole lowpass filters are relatively simple, the overall signal path inside a camera can be derived analytically. Thus the analytically simulated Guassian-lowpass signal can be used to determine the exact timing of the level crossings with arbitrary timing accuracy. Unlike the “video-to-event” event simulators that are inherently frame-based simulation of events [21, 22], analytical generation of events do not suffer from interpolation artifacts, erroneous optical flow calculations, etc. that are particularly severe for unresolved targets like seed particles.

Same technique can be used to generate a synthetic frame. Because the intensity profile of a particle at a particular pixel is known analytically to be Gaussian, we can calculate the exact integration over the exposure time without any temporal interpolation that would fail for unresolved targets such as the seed particles. Examples are shown in Figure 10. Due to the analytical nature of the simulation, there is no theoretical limit to the frame rate or the exposure time (other than the computational resource required to render the simulated events/frames).

The user can specify the simulation experiment setup, such as particle density, the particle size, and flow field; as well as camera parameters such as contrast threshold and one pole lowpass filter in event-based cameras and exposure time and frame rate for frame-based cameras. In our simulation experiment, we chose the particle density and the exposure time to be similar to what is currently used in conventional FB PIV and PTV. The forced vortex with two direction model is chosen as the flow field, since it provides a wide range of velocities with a boundary where two opposite flow direction. The direction of movement is set to be clockwise for all particles located in the center and less than 250-pixel radius while the outer particles are moving in the counter clock manner as demonstrated in figure 11. Table 1 shows the technical details of the simulation model used in our experiment.

For benchmarking, the performance of particle velocimetry is evaluated objectively by the mean square error (MSE) of velocity components (V_x , V_y) and the magnitude, as well as the angle error computed using the Spectral Angel

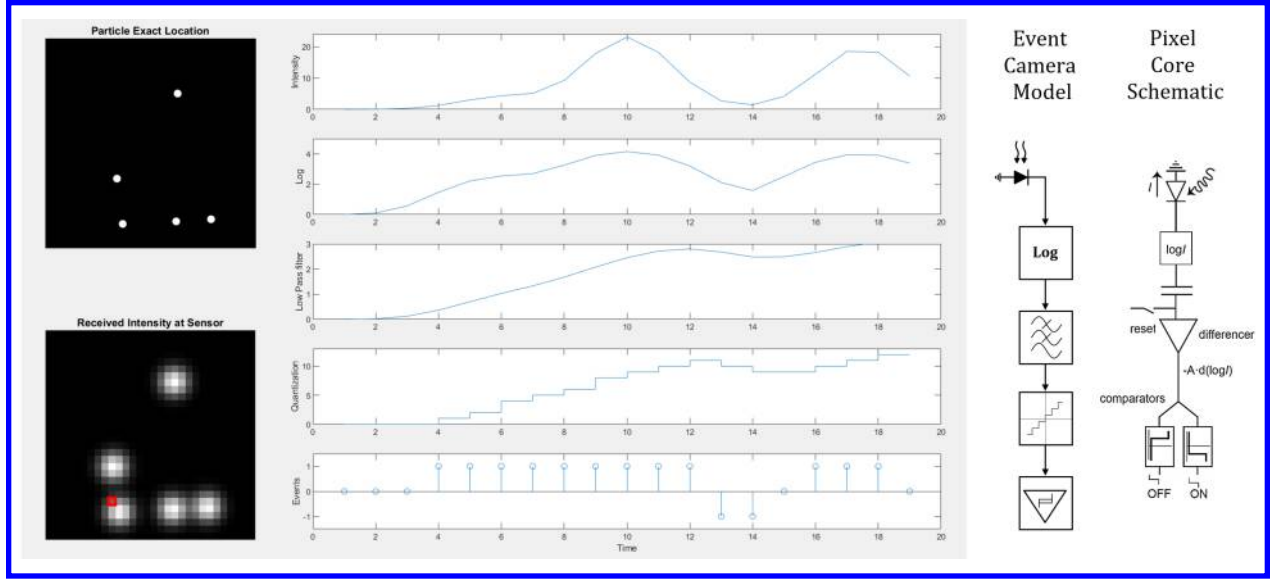


Fig. 10 Generation process of producing the synthetic intensity-based image and the corresponding events mimicking the operation of the physical event-based sensor. (The presented measurement plots demonstrate the change in time of the pixel under investigation (red box) against the intensity and through all steps until producing the events).

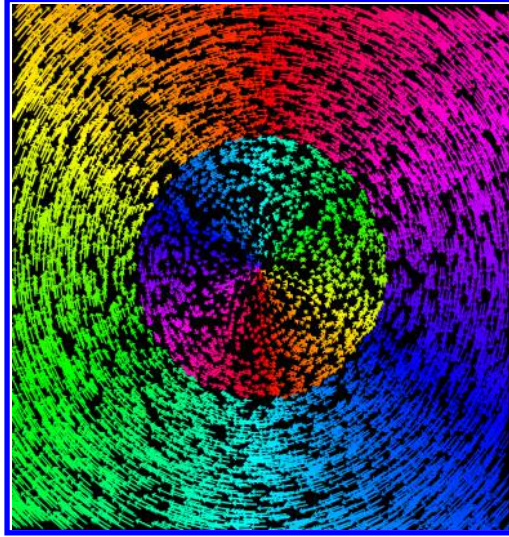


Fig. 11 Ground truth velocity vectors of the synthesise data. The velocity direction is encoded as the color hue.

Mapper (SAM) as described in eq 9:

$$\text{Angle} \left(\hat{\vec{V}}, \vec{V} \right) = \text{Cos}^{-1} \frac{\hat{\vec{V}}^T \vec{V}}{\left\| \hat{\vec{V}} \right\| \left\| \vec{V} \right\|}, \quad (9)$$

where $\vec{V}$ and $\hat{\vec{V}}$ represent ground truth and estimated velocity vectors, respectively. We compare the proposed PEV with EB-PTV methods such as Drazen (2011) [14] and Wang (2020) [15] and EB-PIV method by Willert (2022) [16]. For baseline, recent FB-PIV [3] and FB-PTV [4] methods were considered also. To ensure fair comparison, the parameters for each algorithm were tuned for optimal performance, as summarized in Table 2.

Table 1 Synthesis Model Design Details.

Model Specs	Description
Movement Type	Forced vortex with opposite rotation
Duration	2 (mSec)
Particle Density	0.03 (PPP)
Linear Velocity Range	0 – 50K (Pixel/Sec)
Angular Velocity	0.012 (Degree/uSec)
Particle Size	2 – 4 (Pixel)
Model Sensor Size	501x501 (Pixel)
Intensity Based output	2000 Frame (1 frame/uSec)
Event Based Output	2 mSec of streaming event (x,y,ts,p)

B. Simulation Experiment Result

1. Frame Based PIV and PTV

For the FB-PIV measurement, we used the latest version of PIVlab (particle image velocimetry (PIV) tool with GUI Version 2.62.0.0) by William Thielicke in 2021 [3]. The size of the window and the interrogation time plays an important rule to achieve the best measurements. Larger window size contributes to low spatial resolution while reducing the window size increases the cross-correlation error. Furthermore, the interrogation time must be tuned to achieve the highest temporal accuracy with the minimum relative uncertainty [23, 24] which occurs at 8 -10 pixel displacement of the particles. Therefore, the PIV has been tested with variety of window sizes (16, 24, and 32 pixels) and against wide range of interrogation times (1 to 150 μ Sec).

The FB-PTV tests used Part2Track (A MATLAB package for double frame and time resolved Particle Tracking Velocimetry), which is an open-source system published in 2020 [4]. The system is tested with same range of interrogation times with the minimum exposure time while reporting the accuracy and the number of tracks additionally.

The velocity vectors field in Fig. 12 are produced by PIV and PTV for the optimum case. Even though PIV flow field reconstruction appears to produce a smooth velocity variation, the error stemming from loss of resolution resulted in a poor objective metrics in Table 3. Although FB-PIV outperformed FB-PTV in objective metrics, the situation would reverse with lower particle density (which would be disadvantageous for PIV and advantageous for PTV).

2. Event-Based PIV and PTV

We followed the technical details and replicated the algorithms published in [14–16]. The default settings mentioned in the publications were used to ensure rigorousness (e.g. propagation weight for Wang (2020) [15], mass ration for Drazen (2011) [14], and the voxel sizes (time and space) for the Willert (2022) [16]). Other parameters were optimized experimentally, as reported in Table 2.

Figure 12 shows a visual comparison of the performance of the event-based methods. By comparison, Drazen (2011)[14] has the highest number of tracks but it is unable to handle a wide range of velocities of the particles. Wang (2020) is more accurate than Drazen (2011), but yields fewer number of tracks because of its inability to handle high particle density. The performance of Willert (2022) suffers when the flow field have a wide range of velocities and directions. Even we testing this method with different integration times and window sizes, its performance was the worst in the benchmark. This is reflected in the objective metrics reported in Table 3.

The proposed PEV had stable and most accurate results of all methods. Objectively, it has the lowest MSE and angle error scores of all methods. The number of tracks was the highest among the event-based PTV, albeit fewer than FB-PTV. Visually inspecting Fig. 12, there were almost no outliers in the reconstructed flow field, and the flow boundaries were very cleanly preserved.

V. Real Event-Based Sensor Data Experiments

A. Real Experiment Setup

In addition to the simulation experiments, the proposed PEV was tested in a real-world environment. The particle event velocimetry experiments with the event camera was conducted at the University of Dayton Water Tunnel, shown in Fig. 13(A). The facility is a horizontal free-surface water tunnel with a 6:1 contraction and a 38 cm wide and 46 cm

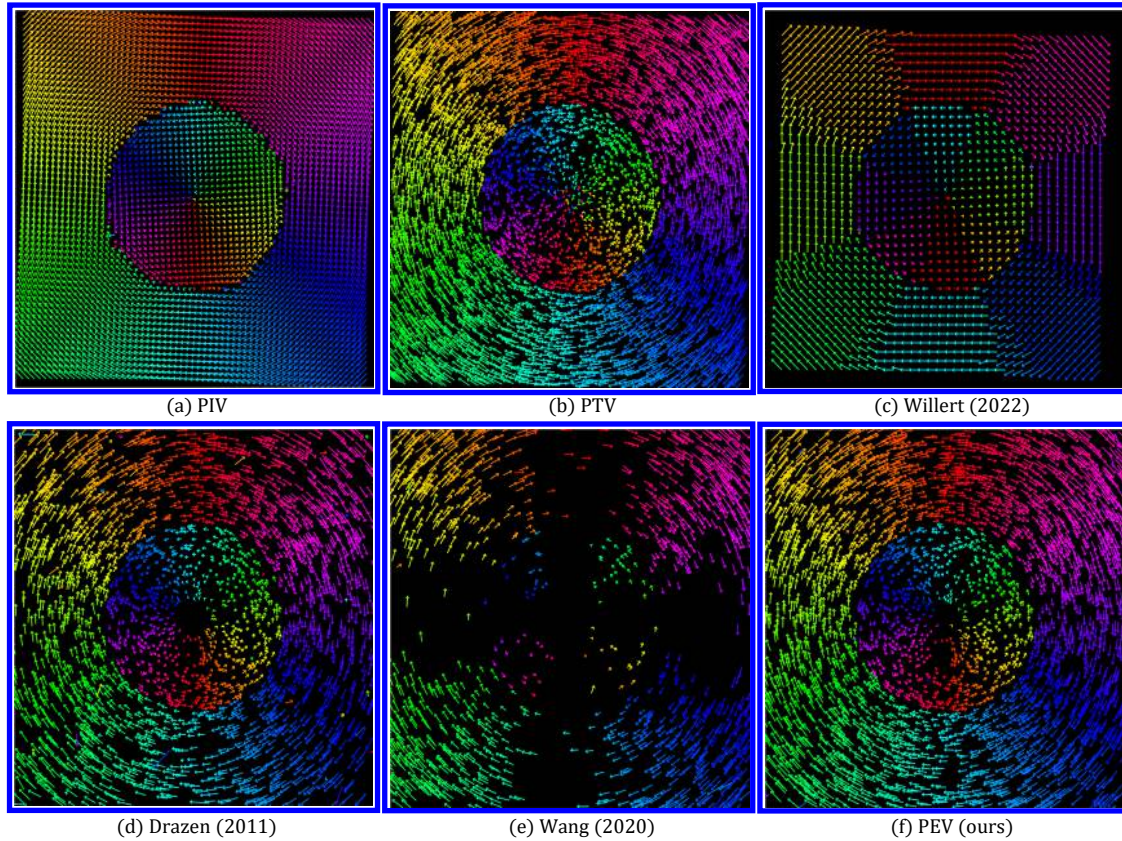


Fig. 12 Velocity vectors of the synthesise data produced using FB and EB methods: (a) Thielicke (2021) [3], (b) Janke (2020) [4], (c) Willert (2022)[16], (d) Drazen (2011)[14], (e) Wang (2020)[15], and (f) PEV(ours).

high test section of 152 cm length. The free-surface speed range is 7.5 cm/s to 38 cm/s. For this study, the freestream velocity of the tunnel was set to 0.28m/s and a flat plate with a 3-inch (0.076m) chord and 6-inch (0.152m) span was used. The tunnel is equipped with a one degree of freedom motion stage consisting of a H2W Technologies linear motor driven by a Xenus XTL-230-18 motor driver from Copley Controls. This motor is outfitted with a rack and pinon gear that converts its linear motion to a rotational motion to pitch a test article in the tunnel test section as shown in Fig. 13(C). The event camera was positioned below the test section focusing on a thin laser plane illuminating the trailing edge of a rotating cylinder with the trailing plate to capture the vortex locking in the wake.

For event-based camera, Prophesee EVK2 camera with 50mm lens was used, yielding a field of view (FOV) of 18x10cm. To the best of authors' knowledge, this is the widest FOV used in EB PIV/PTV experiments reported in the literature thus far. The main difference between the standard 2C2D PIV setup with the conventional PEV system is the that the event camera requires a continuous light laser source to keep track of the particle at 1 MHz resolution.

B. Real Event-Based Sensor Data Experiment Result

As shown in Fig. 14, multiple samples are presented by visualizing the raw events (right) of real experimental data capturing the waves generated by the gust generator along with the resulted velocity vectors produced after performing the PEV system (left). Visually, we can see that the trails of the decaying events match the direction of the velocity vectors. In addition, the smooth variation of the direction and magnitude of the velocity vectors is yield as expected. Moreover, from these results we can see that the PEV system can detect and measure low speed particles located at the center of the vortex as well as the particles with higher velocity at the top. However, in some areas the particle density appears to be less dense than other, which will be investigated during more experimentation and validation which will be in the near future publications.

Table 2 Investigated Parameters for Drazen (2011)[14], Wang (2020)[15], Willert (2022)[16], PTV[4], PIV[3] to obtain best performance for benchmark.

	Method	Parameter	Tested Range	Best Performance
Event Based	Drazen (2011)[14]	Mass weight	0 to 80	.3
	Wang (2020)[15]	Propagation weight	0 to .9	0.4
	Willert (2022)[16]	Window size	24-32-40 pixel	24 pixel
		Exposure time	100 to 1000 μ Sec	1000 μ Sec
Frame Based	PIV[3]	Window size	16-24-32 pixel	16 pixel
		Time between frames	1 to 150 μ Sec	100 μ Sec
	PTV[4]	Time between frames	1 to 250 μ Sec	50 μ Sec

Table 3 Error performance analysis for Drazen (2011)[14], Wang (2020)[15], Willert (2022)[16], PTV[4], PIV[3] against PEV system.

	Method	MSE				# of tracks
		Vx	Vy	Magnitude	Angle	
Event Based	Drazen (2011)[14]	9.885E-04	1.039E-03	8.906E-04	1.86	2540
	Wang (2020)[15]	9.335E-04	9.530E-04	6.069E-04	1.48	862
	Willert (2022)[16]	2.389E-02	2.391E-02	3.658E-02	12.94	N/A
Frame Based	PTV[4]	2.486E-03	2.449E-03	2.384E-03	5.24	4243
	PIV[3]	1.490E-03	1.413E-03	1.278E-03	4.44	N/A
	PEV (OUR)	6.440E-04	6.778E-04	5.451E-04	1.30	3107

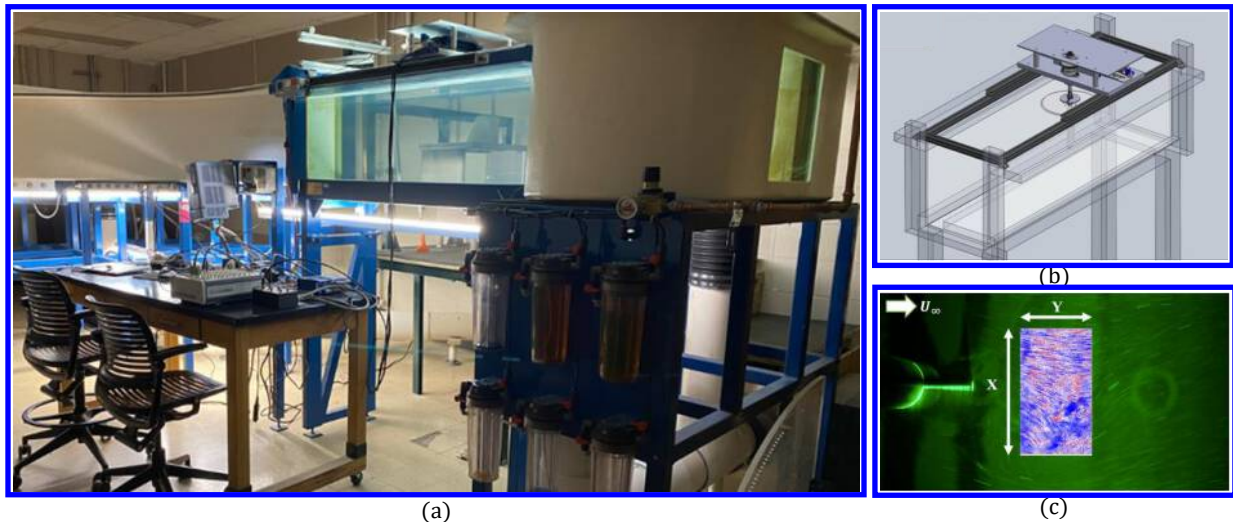


Fig. 13 University of Dayton Water Tunnel facility (a), CAD schematic of gust generator and test article (b), visualization to the FOV (c).

VI. Conclusion

In this article, we proposed a new event-based particle velocimetry technique called particle event velocimetry (PEV). PEV is comprised of pre-processing stages designed to increase robustness to noise, and tracking and regression-based velocity estimation that reconstructs flow field robustly. Compared to the existing FB (PIV [3] and PTV[4]) and EB (Drazen (2011)[14], Wang (2020)[15], Willert (2022)[16]) methods at (0.03 PPP) particle seeding density, PEV improves accuracy in the estimation of velocimetry magnitude, flow direction, and particle track quality.

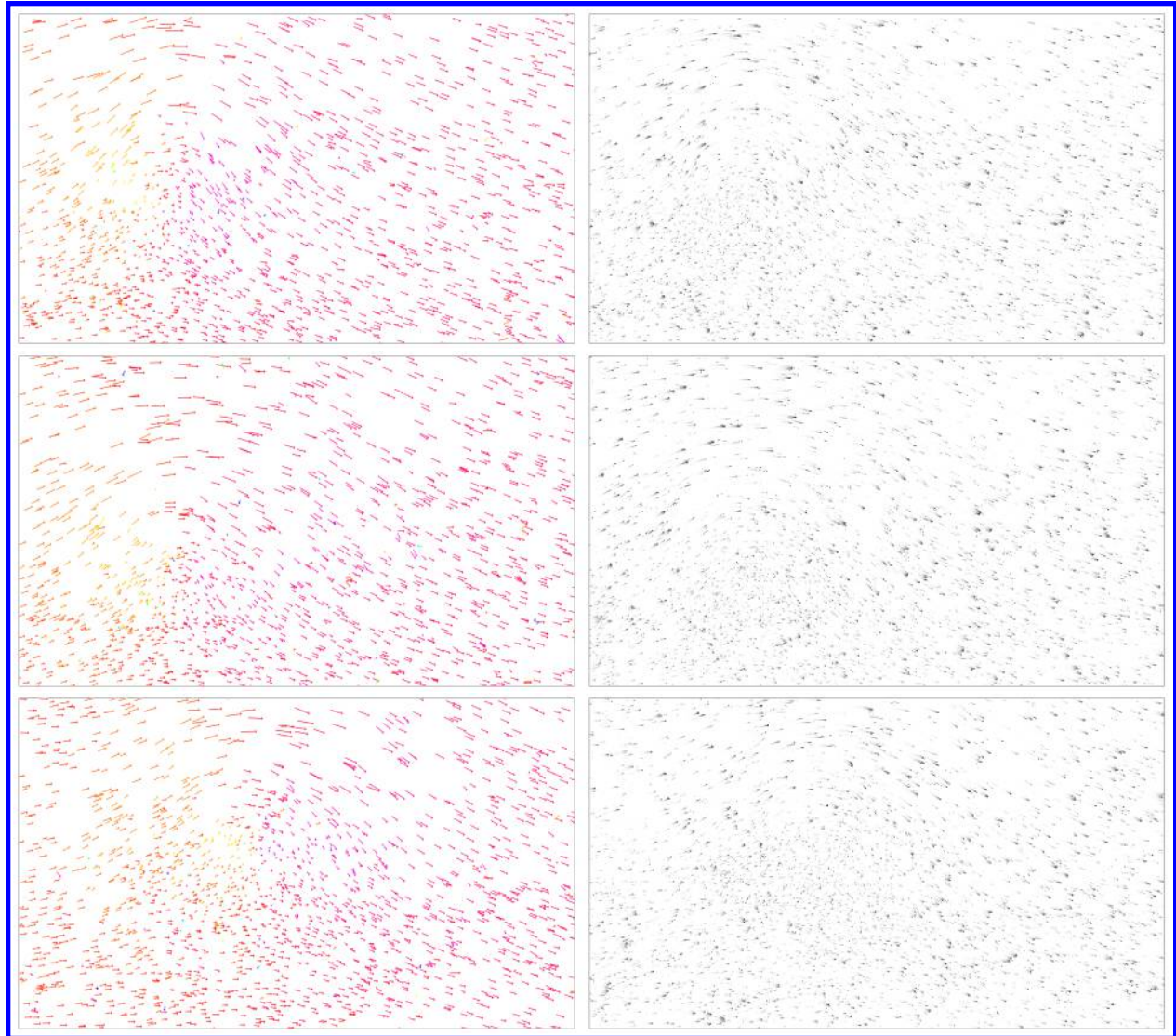


Fig. 14 Samples from a real-world experiment at the Water Tunnel lab. Left: Velocity vector field computed using proposed PEV system where colors indicates the direction of the computed motion. Right: Event data, with time encoded by gray scale (gray=past, black=recent).

Additionally, we developed simulation framework for objectively benchmarking event-based particle velocimetry. This overcomes the limitations of frame-based event simulation methods by generating event streams of moving unresolved targets in continuous time using analytical models of the underlying signal. In simulation, we demonstrated PEV's ability to reconstruct a range of particle velocities from 10 to 50k pixels per second, representing a substantial improvement over the state-of-the-art frame-based and event-based PIV/PTV techniques. Real event-based sensor data experiments conducted in the water tunnel shows robustness to noise and errors in the widest field-of-view reported for event-based particle velocimetry experiment. By the simulation and real experiments, we demonstrated that event-based camera PIV/PTV is cost-effective alternative to frame-based PIV/PTV that simplifies experimentation and computation, requiring only a few calibrations beyond setting biases and extracting extrinsic camera parameters. However, both empirical and numerical verification suggest that further algorithm development could help event camera-based particle velocimetry reach its full potential.

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References

- [1] Scharnowski, S., and Kähler, C. J., “Particle image velocimetry-classical operating rules from today’s perspective,” *Optics and Lasers in Engineering*, Vol. 135, 2020, p. 106185.
- [2] Geisler, R., and Schr, A., “Investigation of unsteady flow fields for flow control research by means of Particle Image Velocimetry,” *Journal of Flow Control, Measurement & Visualization*, Vol. 2014, 2014.
- [3] Thielicke, W., and Sonntag, R., “Particle Image Velocimetry for MATLAB: Accuracy and enhanced algorithms in PIVlab,” *Journal of Open Research Software*, Vol. 9, No. 1, 2021.
- [4] Janke, T., Schwarze, R., and Bauer, K., “Part2Track: A MATLAB package for double frame and time resolved particle tracking velocimetry,” *SoftwareX*, Vol. 11, 2020, p. 100413.
- [5] Qureshi, M. H., Tien, W.-H., and Lin, Y.-J. P., “Performance comparison of particle tracking velocimetry (PTV) and particle image velocimetry (PIV) with long-exposure particle streaks,” *Measurement Science and Technology*, Vol. 32, No. 2, 2020, p. 024008.
- [6] Lichtsteiner, P., and Posch, C., “C. and T. Delbruck, “An 128× 128 120 dB 15 μ s latency temporal contrast vision sensor,” *IEEE Journal of Solid State Circuits*, Vol. 43, 2008, pp. 566–576.
- [7] Gallego, G., Delbrück, T., Orchard, G., Bartolozzi, C., Taba, B., Censi, A., Leutenegger, S., Davison, A. J., Conrads, J., Daniilidis, K., et al., “Event-based vision: A survey,” *IEEE transactions on pattern analysis and machine intelligence*, Vol. 44, No. 1, 2020, pp. 154–180.
- [8] Finatou, T., Niwa, A., Matolin, D., Tsuchimoto, K., Mascheroni, A., Reynaud, E., Mostafalu, P., Brady, F., Chotard, L., LeGoff, F., et al., “5.10 a 1280× 720 back-illuminated stacked temporal contrast event-based vision sensor with 4.86 μ m pixels, 1.066 GEPS readout, programmable event-rate controller and compressive data-formatting pipeline,” *2020 IEEE International Solid-State Circuits Conference-(ISSCC)*, IEEE, 2020, pp. 112–114.
- [9] Ni, Z., Pacoret, C., Benosman, R., Ieng, S., and RÉGNIER*, S., “Asynchronous event-based high speed vision for microparticle tracking,” *Journal of microscopy*, Vol. 245, No. 3, 2012, pp. 236–244.
- [10] Benosman, R., Ieng, S.-H., Clercq, C., Bartolozzi, C., and Srinivasan, M., “Asynchronous frameless event-based optical flow,” *Neural Networks*, Vol. 27, 2012, pp. 32–37.
- [11] Rueckauer, B., and Delbruck, T., “Evaluation of event-based algorithms for optical flow with ground-truth from inertial measurement sensor,” *Frontiers in neuroscience*, Vol. 10, 2016, p. 176.
- [12] Brosch, T., Tschene, S., and Neumann, H., “On event-based optical flow detection,” *Frontiers in neuroscience*, Vol. 9, 2015, p. 137.
- [13] Rebecq, H., Gallego, G., Mueggler, E., and Scaramuzza, D., “EMVS: Event-based multi-view stereo—3D reconstruction with an event camera in real-time,” *International Journal of Computer Vision*, Vol. 126, No. 12, 2018, pp. 1394–1414.
- [14] Drazen, D., Lichtsteiner, P., Häfliger, P., Delbrück, T., and Jensen, A., “Toward real-time particle tracking using an event-based dynamic vision sensor,” *Experiments in Fluids*, Vol. 51, No. 5, 2011, pp. 1465–1469.
- [15] Wang, Y., Idoughi, R., and Heidrich, W., “Stereo event-based particle tracking velocimetry for 3d fluid flow reconstruction,” *European Conference on Computer Vision*, Springer, 2020, pp. 36–53.
- [16] Willert, C. E., and Klinger, J., “Event-based imaging velocimetry: an assessment of event-based cameras for the measurement of fluid flows,” *Experiments in Fluids*, Vol. 63, No. 6, 2022, pp. 1–20.
- [17] Baldwin, R. W., Almatrafi, M., Kaufman, J. R., Asari, V., and Hirakawa, K., “Inceptive event time-surfaces for object classification using neuromorphic cameras,” *Image Analysis and Recognition: 16th International Conference, ICIAR 2019, Waterloo, ON, Canada, August 27–29, 2019, Proceedings, Part II 16*, Springer, 2019, pp. 395–403.
- [18] Willert, C. E., “Event-based imaging velocimetry using pulsed illumination,” *Experiments in Fluids*, Vol. 64, No. 5, 2023, p. 98.

- [19] Gallego, G., Delbruck, T., Orchard, G., Bartolozzi, C., Taba, B., Censi, A., Leutenegger, S., Davison, A. J., Conradt, J., Daniilidis, K., and Scaramuzza, D., “Event-Based Vision: A Survey,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, Vol. 44, No. 1, 2022, pp. 154–180. <https://doi.org/10.1109/TPAMI.2020.3008413>.
- [20] Killick, R., Fearnhead, P., and Eckley, I. A., “Optimal detection of changepoints with a linear computational cost,” *Journal of the American Statistical Association*, Vol. 107, No. 500, 2012, pp. 1590–1598.
- [21] Rebecq, H., Gehrig, D., and Scaramuzza, D., “ESIM: an Open Event Camera Simulator,” *Conf. on Robotics Learning (CoRL)*, 2018.
- [22] Shah, S., Dey, D., Lovett, C., and Kapoor, A., “AirSim: High-Fidelity Visual and Physical Simulation for Autonomous Vehicles,” *Field and Service Robotics*, 2017. URL <https://arxiv.org/abs/1705.05065>.
- [23] Adrian, R. J., “Dynamic ranges of velocity and spatial resolution of particle image velocimetry,” *Measurement Science and Technology*, Vol. 8, No. 12, 1997, pp. 1393–1398. <https://doi.org/10.1088/0957-0233/8/12/003>, URL <https://doi.org/10.1088/0957-0233/8/12/003>.
- [24] Raffel, M., Willert, C. E., Scarano, F., Kähler, C. J., Wereley, S. T., and Kompenhans, J., *Particle Image Velocimetry: A Practical Guide*, Springer, 2018.